

## Project Report: Supermarket Sales Analysis

This project focuses on the analysis of a supermarket sales dataset using Excel, SQL and Power BI to generate actionable insights for business growth.

### Dataset Description

The dataset consisted of **1,000 transactions** and included the following key fields:

- **Invoice\_ID** – Unique identifier of sales transactions
- **Branch & City** – Location of the transaction
- **Customer\_type & Gender** – Customer demographics
- **Product\_line** – Product category purchased
- **Unit\_price, Quantity, COGS, Total, Tax, Gross Income** – Financial metrics
- **Payment\_type** – Mode of payment (Cash, E-wallet, Credit Card)
- **Date & Time** – Transaction time details
- **Rating** – Customer feedback score

### A. Data Transformation and Cleaning using Excel

The data was transformed, cleaned, standardized normalized by changing the date datatype format to the right date datatype format.

### B. Data Extraction and Analysis using

#### SQL SQL Objectives:

1. Calculate the average purchase amount, total sales, and number of purchases for each combination of Customer\_type and gender

Answer:

Members spend more per purchase, especially female members.

Female customers generate more sales overall.

Normal males buy more often, but spend the least per transaction.

2. Select all information from the table

Answer:

All columns were generated from the table

3. Calculate the average quantity sold per transaction for each Product\_line, Identify high-demand products by setting a threshold on this average quantity.

Answer:

Customers buy more than 5 items per transaction in categories like electronic accessories, home and lifestyle, and health and beauty, making them high-demand products.

Electronic accessories lead in average quantity sold, showing strong customer interest.

4. Extract Branch and Rating distribution to get branch with the highest rating

Answer:

Branch A has the highest total customer rating, followed by C and then B.

This suggests that Branch A provides the best overall customer experience and may have better service or environment.

It can serve as a model for improving other branches.

5. Identify Product\_line with the highest unit price

Answer:

Fashion accessories have the highest total unit price, making it the most valuable product line in terms of pricing.

This suggests it contributes significantly to revenue and should be promoted or featured more to boost profit.

### c. Supermarket Sales visualization using Power BI

The main objectives of the project were:

- To **track key performance indicators (KPIs)** such as revenue, cost, profit, gross margin, transactions, and customer satisfaction ratings.
- To **analyze sales trends** across time, branches, cities, and product lines.
- To **understand customer behavior** by demographics, purchase methods, and membership type.
- To create an **interactive dashboard** that supports decision-making and business strategy.

**Methodology:** The following steps were undertaken to build the dashboard:

1. **Data Cleaning & Transformation**
  - Verified data types (dates, numbers, categorical fields).
  - Created calculated columns for Month, Day, and Hour of transaction.
  - Built a Date Table for time intelligence calculations (YTD, MTD, trends).

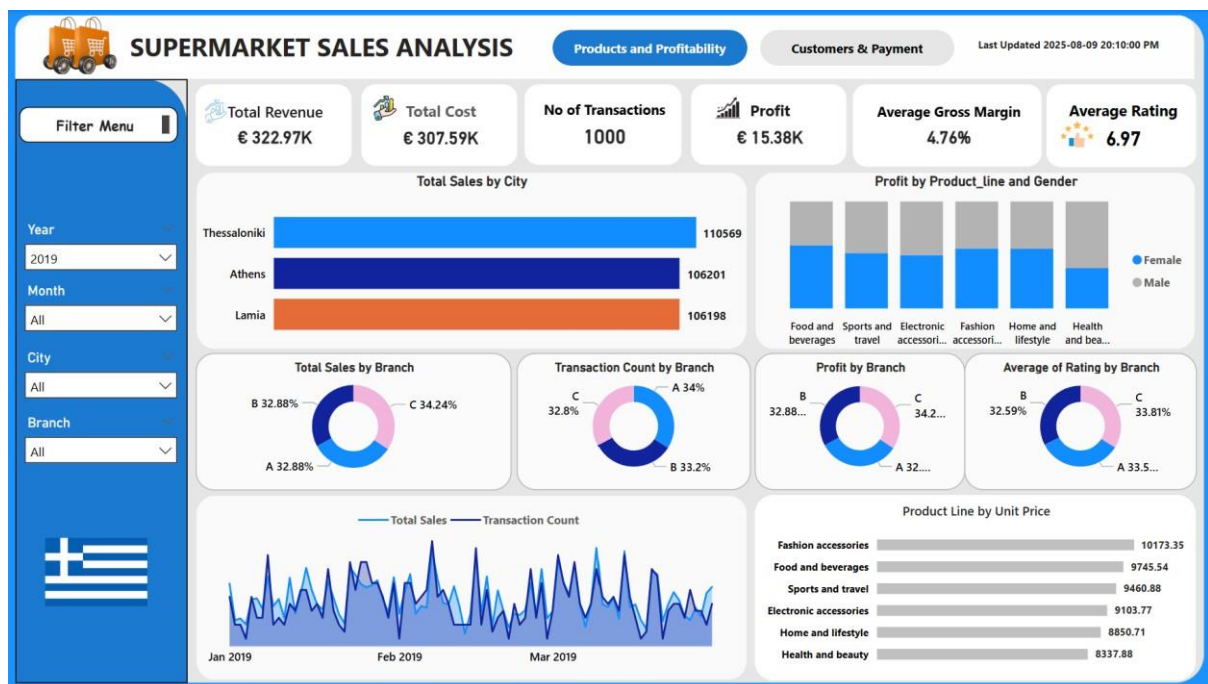
## 2. DAX Measures

- Total Sales (Revenue), Total Cost (COGS), Gross Profit, Gross Margin %
- Transactions, Average Order Value, Average Basket Size
- Sales by Branch, Product Line, Payment Type, and Customer Type
- Average Rating

## 3. Dashboard Design

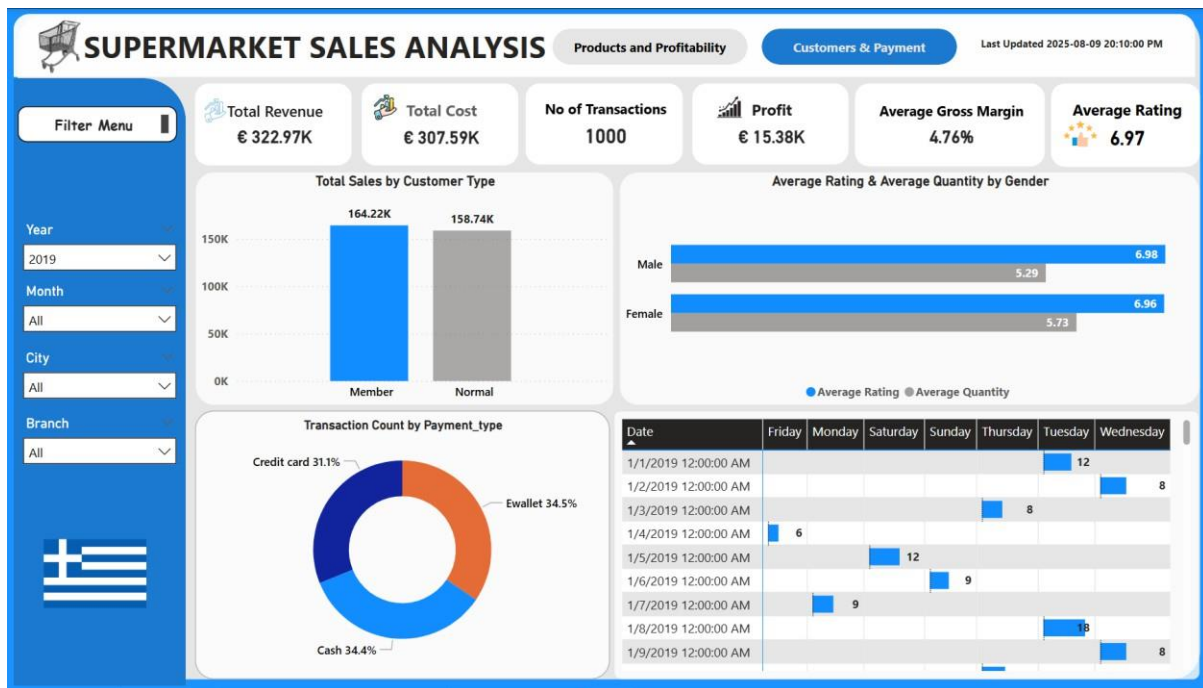
### Page 1: Products & Profitability

- Revenue, Cost, Transactions, Profit, Gross Margin, and Customer Rating as KPI cards.
- Branch performance comparisons (Sales, Profit, Ratings).
- Product line profitability breakdown by gender.
- Monthly sales trend visualization.
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### Page 2: Customers & Payments

- Sales by Customer Type and Gender.
- Transaction count by Payment Method.
- Heatmap of sales across weekdays and dates.
- Rating and purchase quantity comparison between Male and Female customers.



## Key Insights

### Financial Performance

- Total Revenue: €322.97K with Gross Profit of €15.38K, resulting in an Average Gross Margin of 4.76%.
- Total Cost: €307.59K indicates tight margins, emphasizing the need for better pricing or cost optimization.

### Branch & City Analysis

- Sales are fairly balanced across Thessaloniki, Athens, and Lamia, each contributing over €100K.
- Branch C led in total sales and profit share (~34%).

### Product Performance

- Food and Beverages and Sports and Travel were among the top-performing product lines.
- Gender-based profitability showed male customers contributed slightly more to profitability across product lines.

### Customer Behavior

- Members generated slightly more revenue than Normal customers, highlighting the value of loyalty programs.

- Customer satisfaction remained strong with an average rating of 6.97/10, though improvement opportunities exist.

### **Payment Preferences**

- Payment was evenly distributed: E-wallet (34.5%), Cash (34.4%), and Credit Card (31.1%). This indicates a healthy adoption of digital payments.

### **Temporal Trends**

Transaction activity was consistent across January–March 2019, with notable spikes on specific days of the week (especially Tuesdays and Fridays).

## **RECOMMENDATIONS**

1. Explore renegotiating supplier contracts or revising product pricing for higher-margin categories.
2. Since Members spend more than Normal customers, expanding membership benefits can boost retention.
3. Encourage e-wallet use through discounts or cashback campaigns, as it is already the most popular payment method.
4. Higher sales on specific weekdays suggest opportunities for targeted campaigns and staffing adjustments.
5. While ratings are decent, initiatives such as better after-sales service or quality improvement could lift customer satisfaction.

## **CONCLUSION**

This Power BI Dashboard provides a comprehensive view of supermarket performance, highlighting sales trends, customer behavior, and financial health. By leveraging these insights, management can take data-driven decisions to boost profitability, enhance customer loyalty, and optimize operations.

The dashboard demonstrates how business intelligence tools like Power BI can transform raw transaction data into strategic insights for growth and competitiveness.